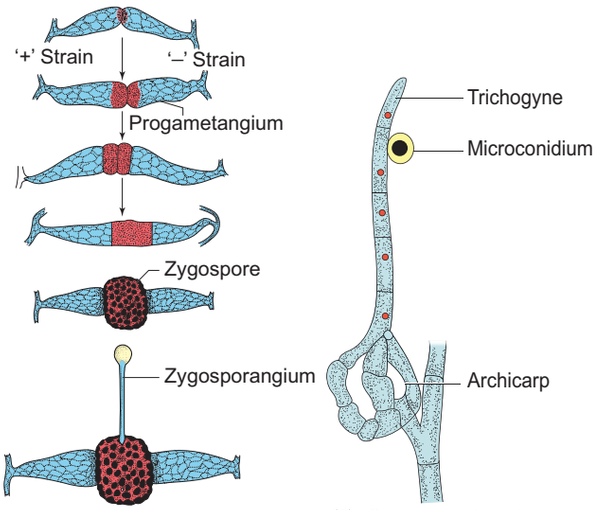


(h) Gametangial copulation - *Rhizopus*

(i) Spermatisation - *Neurospora*

Figure 1.19: Reproduction in Fungi

1.5.4 Classification of Fungi

Many mycologists have attempted to classify fungi based on vegetative and reproductive characters. Traditional classifications categorise fungi into 4 classes – Phycomycetes, Ascomycetes, Basidiomycetes and Deuteromycetes. Among these 'Phycomycetes' include fungal species of Oomycetes, Chytridiomycetes and Zygomycetes which are considered as lower fungi indicating algal origin of fungi. Constantine J. Alexopoulos and Charles W. Mims in the year 1979 proposed the classification of fungi in the book entitled 'Introductory Mycology'. They classified fungi into three divisions namely Gymnomycota, Mastigomycota and Amastigomycota. There are 8 subdivisions, 11 classes, 1 form class and 3 form subclasses in the classification proposed by them.

1.5.5 Kingdom : Myceteae (Fungi)

These include achlorophyllous, saprophytic or parasitic organisms with unicellular or multicellular (Mycelium) thallus surrounded by chitinous cell wall. Nutrition is absorptive except slime molds.Reproduction is through asexual and sexual methods.

Division : I Gymnomycota

Nutrition is phagotrophic, members of this group lack cell wall. Example. *Dictyostelium*

Division :II Mastigomycota

Flagellate cells are present (Gamete/ Zoospore). Nutrition absorptive, mycelium coenocytic. Example : *Albugo.*

Division : III Amastigomycota

Unicellular to multicellular forms are included. The mycelium is septate. Asexual reproduction occurs by budding,

The outline of the classification is given below:

KINGDOM - MYCETEAE

DIVISION – GYMNOMYCOTA

SUBDIVISION – 1
ACRASIOGYMNOMYCOTINA
CLASS – Acrasiomycetes

SUBDIVISION – 2
PLASMODIOGYMNOMYCOTINA
CLASS – Protosteliomycetes
CLASS- Myxomycetes

DIVISION - MASTIGOMYCOTA

SUBDIVISION – 1
HAPLOMASTIGOMYCOTINA
CLASS – Chytridiomycetes
CLASS – Hyphochytridiomycetes
CLASS – Plasmodiophoromycetes

SUBDIVISION – 2
DIPLOMASTIGOMYCOTINA
CLASS - Oomycetes

DIVISION – AMASTIGOMYCOTA

SUBDIVISION -1
ZYGOMYCOTINA
Class – Zygomycetes
Class- Trichomycetes

SUBDIVISION -2
ASCOMYCOTINA
CLASS – Ascomycetes

SUBDIVISION -3
BASIDIOMYCOTINA
CLASS – Basidiomycetes

SUBDIVISION -4
DEUTEROMYCOTINA
CLASS - Deuteromycetes

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